

ACTIVITY PERTEMUAN 5

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KELAS : 1IA04

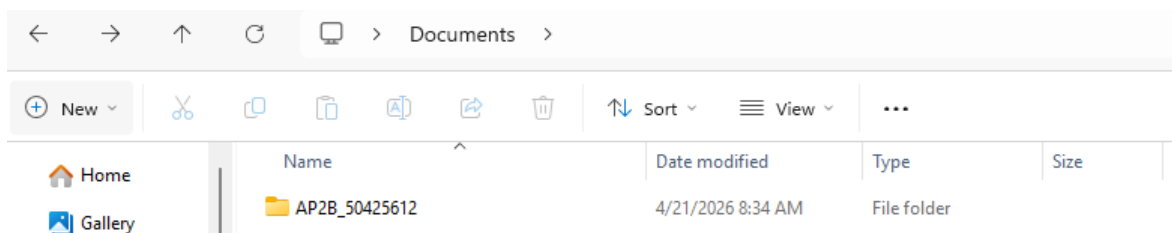
MATERI : API WITH FLASK

MATA PRAKTIKUM : ALGORITMA PEMROGRAMAN 2B

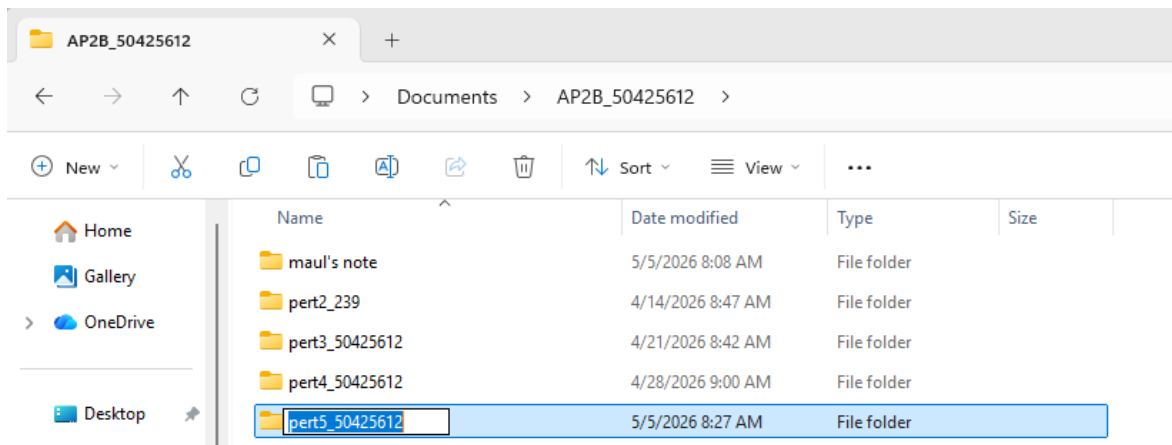
(Screenshoot langkah-langkah sesuai video pembelajaran dan jelaskan dengan ringkas)

1. Persiapan Struktur Folder

- Masuk ke folder utama praktikum, yaitu **AP2B_50425612**.



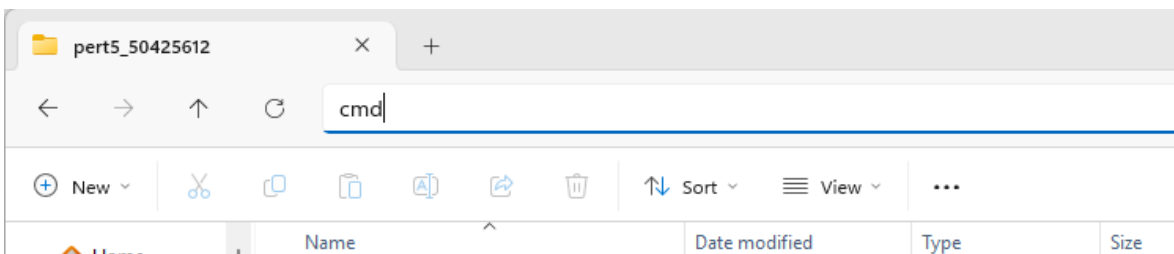
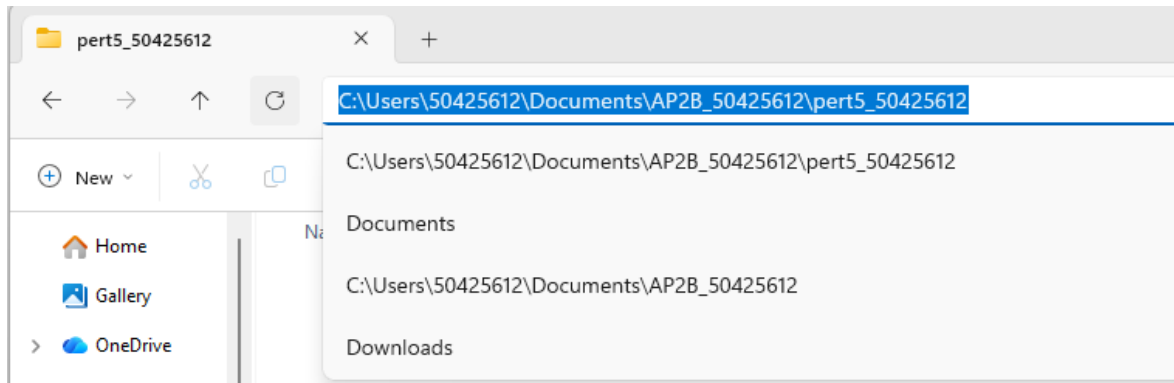
- Membuat sub-folder baru dengan format penamaan **pert5_50425612** sebagai wadah file proyek pertemuan ini.



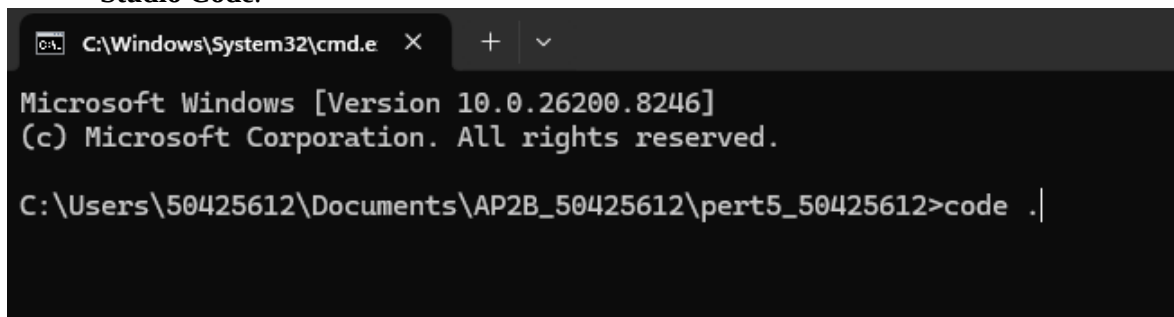
- Logikanya, pemisahan folder ini dilakukan agar manajemen file antar pertemuan tetap rapi dan terstruktur.

2. Membuka VS Code

- Klik pada *address bar* folder yang telah dibuat, ketik **cmd**, lalu tekan Enter untuk membuka Command Prompt pada direktori tersebut.

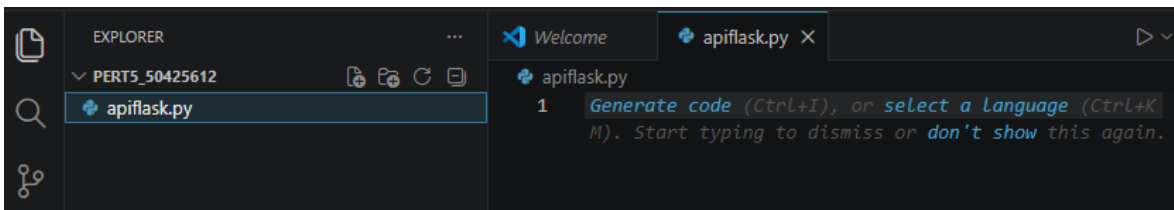


- Ketik perintah code . untuk membuka folder tersebut secara otomatis ke dalam **Visual Studio Code**.



3. Setup file dan buat program

- Di dalam VS Code, buat file Python baru dengan nama **apiflask.py**.



- Copy paste full code yang sudah di buat kemarin pada praktikum pertemuan 4 yang

bernama **app.py** ke dalam file **apiflask.py** praktikum pertemuan M5 ini

```
app.py
app.py > format_rupiah
1  from flask import Flask, jsonify, request
2  import json
3  app = Flask(__name__)
4  datas = [
5      {
6          'id': 1,
7          'items': [
8              { 'title': 'Madilog',
9                'author': 'Tan Malaka',
10               'price': 84000
11             }
12          ]
13      }, {
14          'id': 2,
15          'items': [
16              {
17                  'title': 'Animal Farm',
18                  'author': 'George Orwell',
19                  'price': 50000
20              }
21          ]
22      }
23  ]
24  def format_rupiah(amount):
25      return f"Rp{amount:,.0f}".replace(",", ".")
```

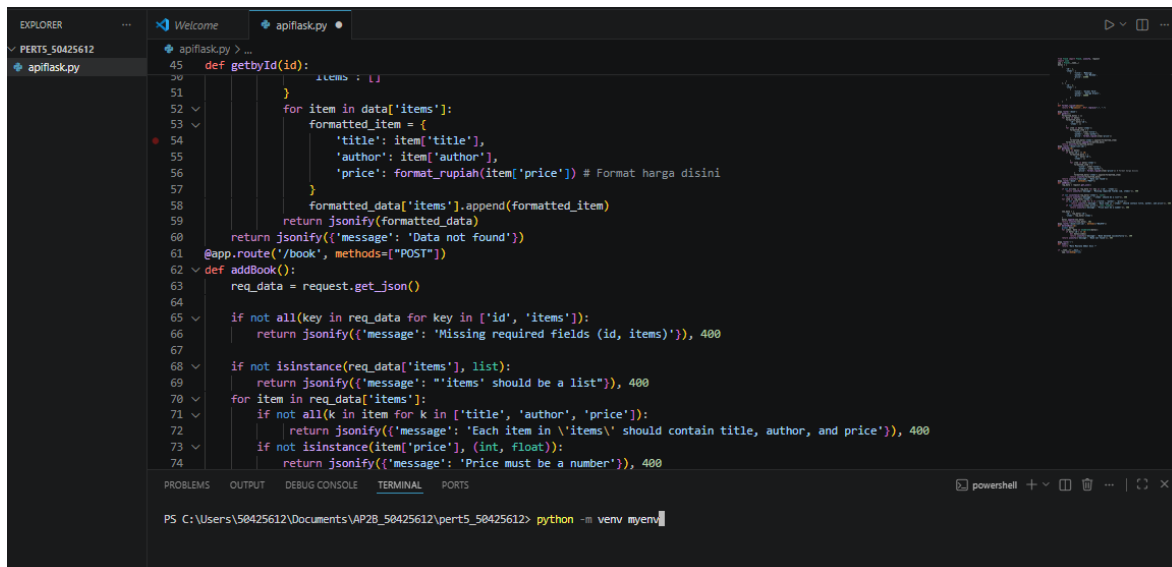
```

27 @app.route('/book')
28 def getAll():
29     formatted_datas = []
30     for data in datas:
31         formatted_data = {
32             'id': data['id'],
33             'items': []
34         }
35         for item in data['items']:
36             formatted_item = {
37                 'title': item['title'],
38                 'author': item['author'],
39                 'price': format_rupiah(item['price'])
40             }
41             formatted_data['items'].append(formatted_item)
42         formatted_datas.append(formatted_data)
43     return jsonify(formatted_datas)
44 @app.route('/book/<int:id>')
45 def getbyId(id):
46     for data in datas:
47         if data['id'] == id:
48             formatted_data = {
49                 'id': data['id'],
50                 'items': []
51             }
52             for item in data['items']:
53                 formatted_item = {
54                     'title': item['title'],
55                     'author': item['author'],
56                     'price': format_rupiah(item['price']) # Format harga disini
57                 }
58                 formatted_data['items'].append(formatted_item)
59             return jsonify(formatted_data)
60     return jsonify({'message': 'Data not found'})
61 @app.route('/book', methods=["POST"])
62 def addBook():
63     req_data = request.get_json()
64
65     if not all(key in req_data for key in ['id', 'items']):
66         return jsonify({'message': 'Missing required fields (id, items)'}), 400
67
68     if not isinstance(req_data['items'], list):
69         return jsonify({'message': "'items' should be a list"}), 400
70     for item in req_data['items']:
71         if not all(k in item for k in ['title', 'author', 'price']):
72             return jsonify({'message': 'Each item in \'items\' should contain title, author, and price'}), 400
73         if not isinstance(item['price'], (int, float)):
74             return jsonify({'message': 'Price must be a number'}), 400
75
76     new_data = {
77         'id': req_data['id'],
78         'items': req_data['items']
79     }
80     datas.append(new_data)
81     return jsonify(new_data), 201
82 @app.route('/book/<int:id>', methods=["DELETE"])
83 def deleteBook(id):
84     global datas
85     for index, data in enumerate(datas):
86         if data['id'] == id:
87             del datas[index]
88             return jsonify({'message': 'Book deleted successfully'}), 200
89     return jsonify({'message': 'Book not found'}), 404

```

4. Konfigurasi Lingkungan Virtual (Venv), instalasi flask, dan aktivasi server

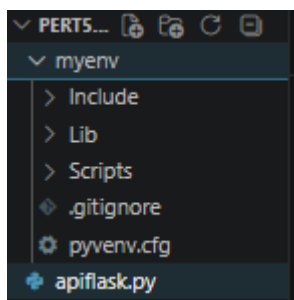
- Buka terminal internal VS Code (Ctrl + Shift + ') dan ketik perintah `python -m venv .myenv` untuk membuat *virtual environment* baru.



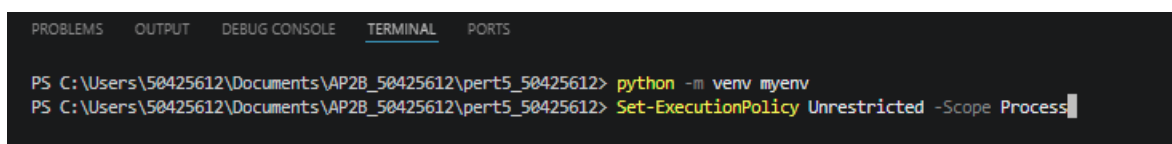
```
45 def getbyId(id):
46     items = []
47     for item in data['items']:
48         formatted_item = {
49             'title': item['title'],
50             'author': item['author'],
51             'price': format_rupiah(item['price']) # Format harga disini
52         }
53         formatted_data['items'].append(formatted_item)
54     return jsonify(formatted_data)
55     return jsonify({'message': 'Data not found'})
56 @app.route('/book', methods=['POST'])
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61     if not isinstance(req_data['items'], list):
62         return jsonify({'message': '"items" should be a list'}), 400
63     for item in req_data['items']:
64         if not all(k in item for k in ['title', 'author', 'price']):
65             return jsonify({'message': 'Each item in \'items\' should contain title, author, and price'}), 400
66         if not isinstance(item['price'], (int, float)):
67             return jsonify({'message': 'Price must be a number'}), 400
```

PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> python -m venv myenv

- File baru bernama myenv sudah tergenerate adalah tanda bahwa myenv sudah dijalankan.

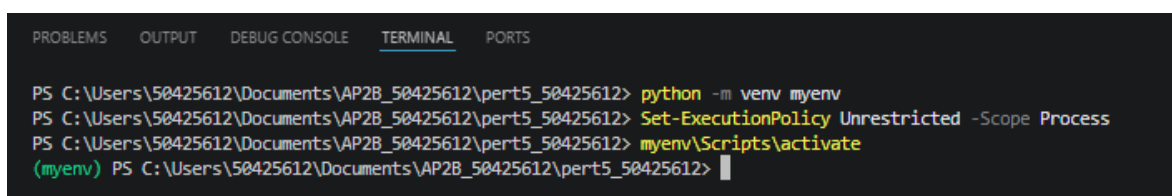


- Atur kebijakan eksekusi terminal dengan perintah Set-ExecutionPolicy Unrestricted -Scope Process agar skrip aktivasi dapat dijalankan.



```
PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> python -m venv myenv
PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> Set-ExecutionPolicy Unrestricted -Scope Process
```

- Aktivasi lingkungan dengan mengetik .env\Scripts\activate. Tanda (.env) pada terminal menunjukkan lingkungan sudah aktif.
- Mengaktivasi dengan activate, .myenv adalah tanda program kita sudah berada dalam environmentnya



```
PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> python -m venv myenv
PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> Set-ExecutionPolicy Unrestricted -Scope Process
PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612> myenv\Scripts\activate
(myenv) PS C:\Users\58425612\Documents\AP2B_58425612\pert5_58425612>
```

- Jalankan perintah pip install Flask untuk memasang framework Flask ke dalam lingkungan virtual tersebut.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612> python -m venv myenv
PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612> Set-ExecutionPolicy Unrestricted -Scope Process
PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612> myenv\Scripts\activate
(myenv) PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612> pip install flask
Collecting flask
  Using cached flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Using cached click-8.3.3-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Using cached itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Using cached markupsafe-3.0.3-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Using cached werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Collecting colorama (from click>=8.1.3->flask)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Using cached flask-3.1.3-py3-none-any.whl (103 kB)
Using cached blinker-1.9.0-py3-none-any.whl (8.5 kB)
Using cached click-8.3.3-py3-none-any.whl (110 kB)
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Using cached markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Using cached werkzeug-3.1.8-py3-none-any.whl (226 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: markupsafe, itsdangerous, colorama, blinker, werkzeug, jinja2, click, flask
Successfully installed blinker-1.9.0 click-8.3.3 colorama-0.4.6 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8

[notice] A new release of pip is available: 26.0.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(myenv) PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612>

```

- Jalankan server melalui terminal dengan perintah python apiflask.py.

```

(myenv) PS C:\Users\S0425612\Documents\AP2B_50425612\pert5_50425612> python apiflask.py
* Serving Flask app 'apiflask'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 131-585-823

```

- Setelah muncul pesan Running on <http://127.0.0.1:5000> , copy local link tersebut untuk nanti digunakan berulang kali pada platform POSTMAN.

5. Integrasi POSTMAN

- Buka aplikasi POSTMAN yang sudah terinstall dan klik skip and to to the app



Create an account or sign in

Create Free Account

Sign in

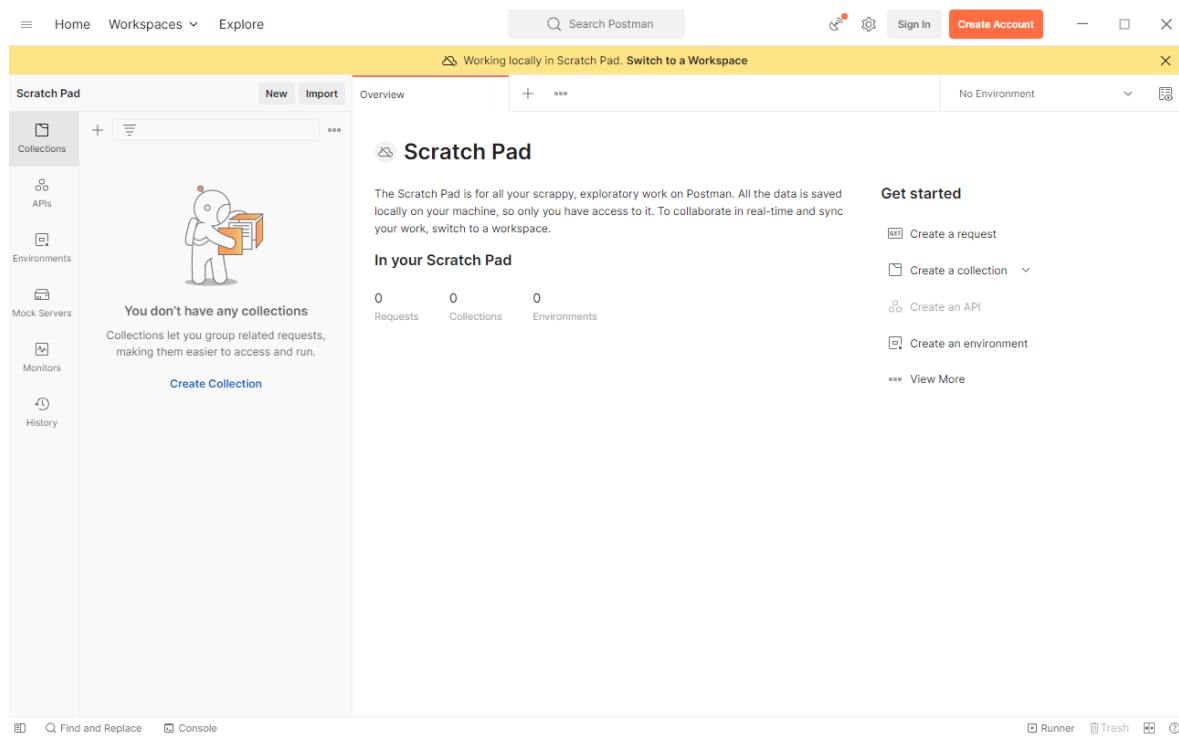
Create your account or sign in later? [Skip and go to the app](#)

A free Postman account lets you

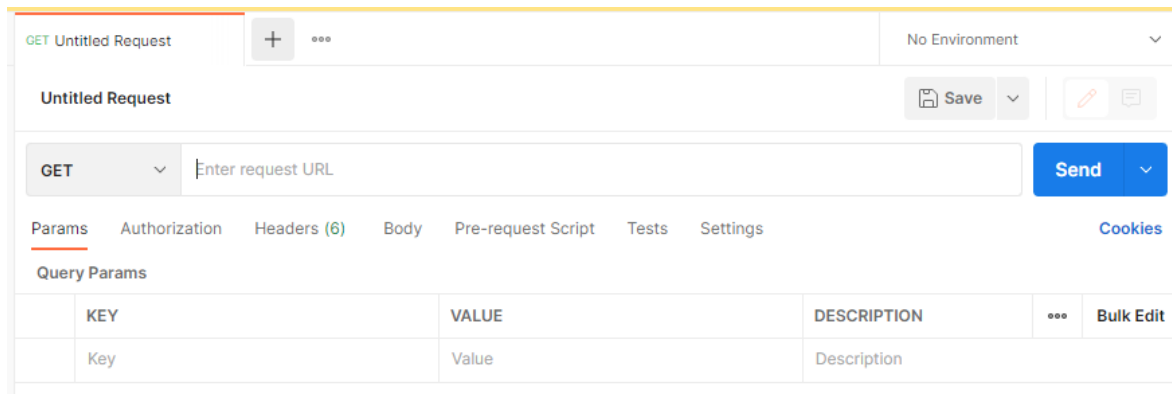
- ✓ Organize all your API development in workspaces
- ✓ Create public workspaces to collaborate with over 10 million developers
- ✓ Back up your work on Postman's cloud
- ✓ Experience the best API development platform for free!



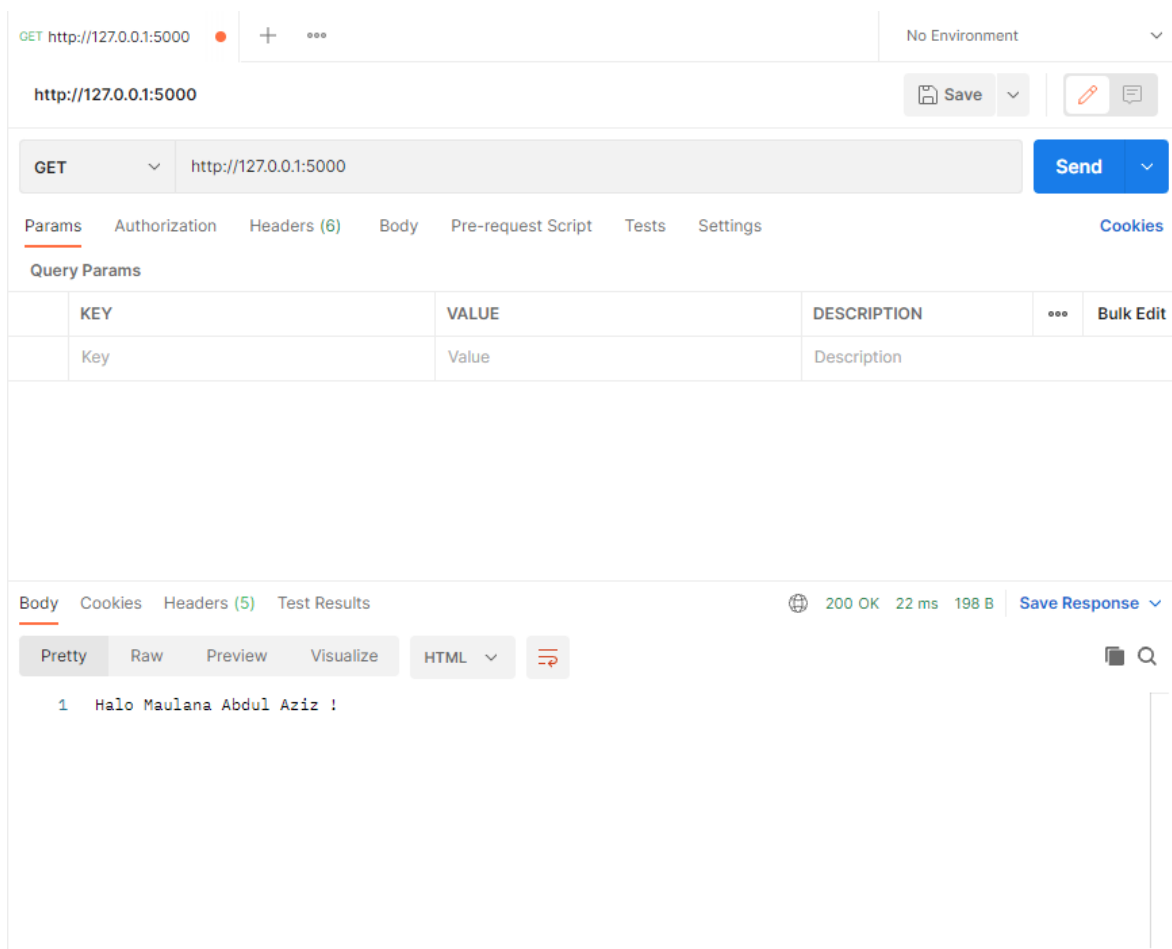
- Berikut tampilan POSTMAN awal



- Klik tanda + pada laman untuk menambahkan tab api editor



- Paste local link yang sudah di copy pada Langkah sebelumnya, maka isi GET pada file program dari dalam vscode akan muncul pada terminal aplikasi POSTMAN ini



- Uji dengan <http://127.0.0.1:5000/book> untuk memverifikasi page GET yang berisi list buku

GET http://127.0.0.1:5000/ + ... No Environment

http://127.0.0.1:5000/book Save ✎ 💬

GET ▼ http://127.0.0.1:5000/book Send ▼

Params Authorization Headers (6) Body Pre-request Script Tests Settings Cookies

Query Params

	KEY	VALUE	DESCRIPTION	...	Bulk Edit
	Key	Value	Description		

Body Cookies Headers (5) Test Results 🌐 200 OK 14 ms 470 B Save Response ▼

Pretty Raw Preview Visualize JSON ▼ ☰

```

1  [
2    {
3      "id": 1,
4      "items": [
5        {
6          "author": "Tan Malaka",
7          "price": "Rp84.000",
8          "title": "Madilog"
9        }
10     ]
11  },
12  {
13    "id": 2,
14    "items": [
15      {
16        "author": "George Orwell",
17        "price": "Rp50.000",
18        "title": "Animal Farm"
19      }
20    ]
21  }
22 ]

```

- Buka method baru yaitu POST pada tab, kli body – raw – dan ubah text ke JSON

GET http://127.0.0.1:5000/bi POST http://127.0.0.1:5000/i + ... No Environment

http://127.0.0.1:5000/book Save ✎ 💬

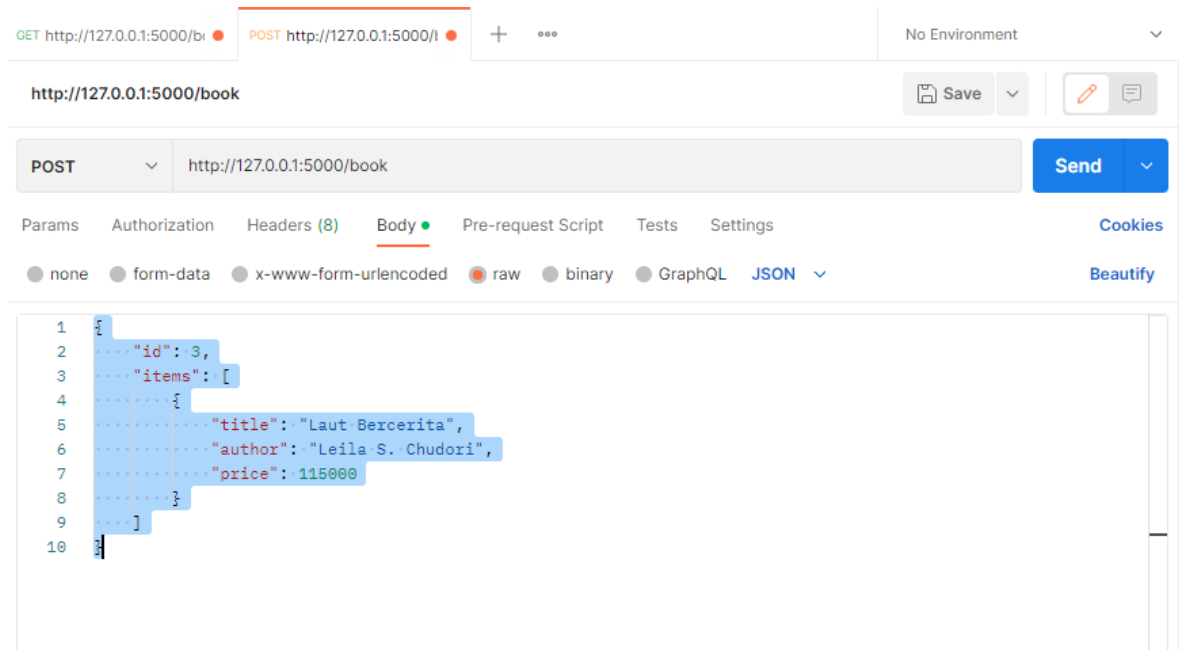
POST ▼ http://127.0.0.1:5000/book Send ▼

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

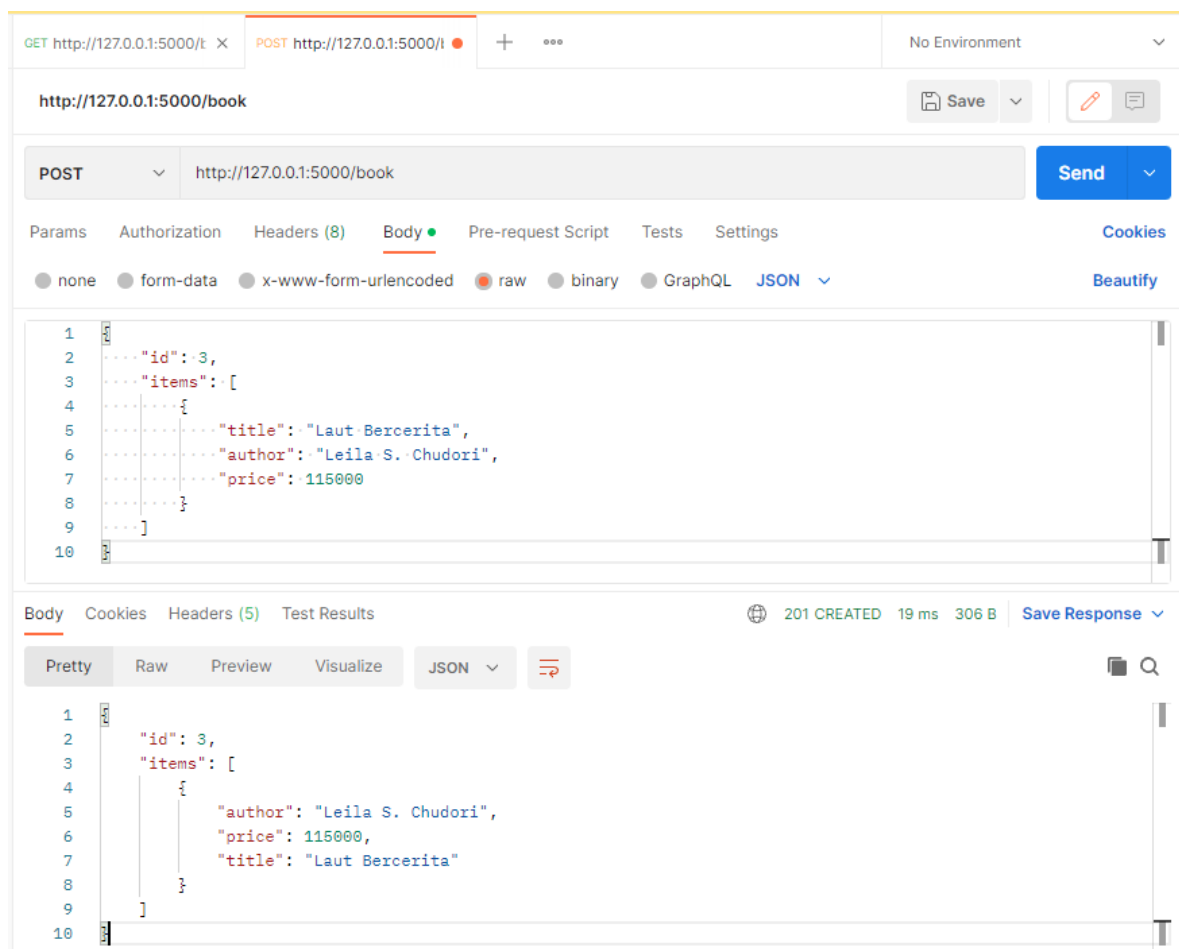
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼ Beautify

1 |

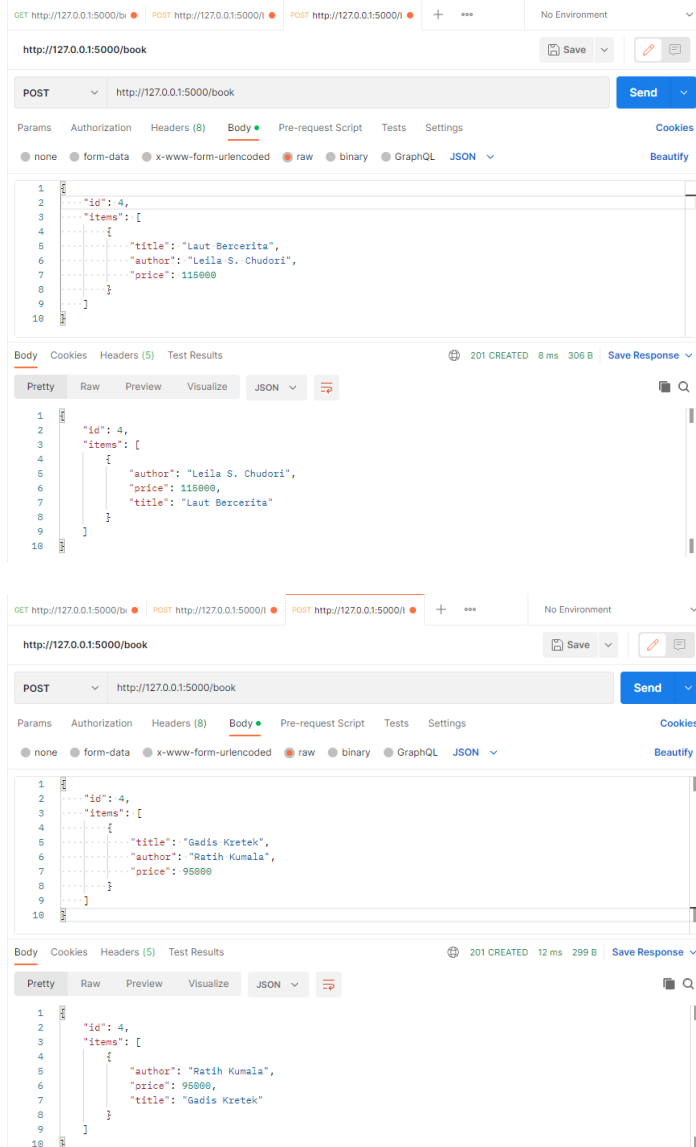
- Tambahkan data buku baru dalam method POST editor terminal tersebut sesuai yang disediakan pada M5 forum satu persatu dari ID: 3



- Klik send, maka terminal output dibawahnya akan muncul sebagai tanda bahwa POST berhasil



- Itu adalah contoh Langkah pada id : 3
- Ulangi Langkah menambahkan method POST sampai ke id : 7Buka aplikasi POSTMAN yang sudah terinstall di perangkat.



GET http://127.0.0.1:5000/ POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book + ... No Environment

http://127.0.0.1:5000/book Save Send

POST http://127.0.0.1:5000/book Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies Beautify

```
1 {
2   "id": 5,
3   "items": [
4     {
5       "title": "The Psychology of Money",
6       "author": "Morgan Housel",
7       "price": 110000
8     }
9   ]
10 }
```

Body Cookies Headers (5) Test Results 201 CREATED 11 ms 312 B Save Response

Pretty Raw Preview Visualize JSON Beautify

```
1 {
2   "id": 5,
3   "items": [
4     {
5       "author": "Morgan Housel",
6       "price": 110000,
7       "title": "The Psychology of Money"
8     }
9   ]
10 }
```

GET http://127.0.0.1:5000/ POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book + ... No Environment

http://127.0.0.1:5000/book Save Send

POST http://127.0.0.1:5000/book Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies Beautify

```
1 {
2   "id": 6,
3   "items": [
4     {
5       "title": "Atomic Habits",
6       "author": "James Clear",
7       "price": 120000
8     }
9   ]
10 }
```

Body Cookies Headers (5) Test Results 201 CREATED 9 ms 300 B Save Response

Pretty Raw Preview Visualize JSON Beautify

```
1 {
2   "id": 6,
3   "items": [
4     {
5       "author": "James Clear",
6       "price": 120000,
7       "title": "Atomic Habits"
8     }
9   ]
10 }
```

GET http://127.0.0.1:5000/ POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book POST http://127.0.0.1:5000/book + ... No Environment

http://127.0.0.1:5000/book Save Send

POST http://127.0.0.1:5000/book Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies Beautify

```
1 {
2   "id": 7,
3   "items": [
4     {
5       "title": "Sapiens",
6       "author": "Yuval Noah Harari",
7       "price": 150000
8     }
9   ]
10 }
```

Body Cookies Headers (5) Test Results 201 CREATED 7 ms 300 B Save Response

Pretty Raw Preview Visualize JSON Beautify

```
1 {
2   "id": 7,
3   "items": [
4     {
5       "author": "Yuval Noah Harari",
6       "price": 150000,
7       "title": "Sapiens"
8     }
9   ]
10 }
```